Harsh Kumar Singh

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LinkedIn | GitHub | LeetCode | CodeChef | GeeksforGeeks

SUMMARY

Computer Engineering undergraduate (Class of 2027) focused on software development, strong in DSA, object-oriented design, DBMS, OS and Computer Networks, with 1000+ solved problems. Two internships (industry and IIT research) building, testing and deploying REST APIs, data pipelines and modular systems at scale.

EDUCATION

National Institute of Advanced Manufacturing Technology (NIAMT, formerly NIFFT), Ranchi **2023 – 2027**
B.Tech in Computer Engineering | **CGPA: 8.46/10** | No backlogs
Class XII (BSEB), Purnea College, Purnea 2022 | 86.0%
Class X (CBSE), Saraswati Vidya Mandir 2020 | 84.4%
Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, AI & Machine Learning

EXPERIENCE

Machine Learning Intern | Evynx Technologies Pvt. Ltd. **Jun 2026 – Aug 2026**
Crowd Forecasting & Tour Package Recommendation System (Live Demo)

- Developed and deployed a FastAPI/Flask backend (11 REST endpoints) on Render, resolving **95.2%** of free-text route stops to known places.
- Automated the data pipeline, cutting storage by **64%** and feature count by **58%** with LightGBM-based feature selection.
- Engineered a four-layer scoring engine fusing festival, weather, search-trend and calendar signals to rank all **632 tour packages** across 12,655 places.
- Reached **92.8%** peak-month forecast accuracy on 291 unseen locations (vs 53.6% baseline), validated against government tourism data.

Research Intern (SRIS 2026) | IIT (ISM) Dhanbad **May 2026 – Jun 2026**
Graph Neural Networks for Recommender Systems | *Guide: Prof. Rajendra Pamula, Dept. of CSE* (Code | Demo)

- Designed and benchmarked **9 GNN architectures** (LightGCN, APPNP, Residual, JK-Net) in a modular PyTorch codebase on datasets of up to **1.13M** interactions.
- Created a reusable embedding module (all-MiniLM-L6-v2) used as a drop-in component across all 9 models.
- Beat all **8 baselines** on Recall@20 and NDCG@20 by fusing semantic review embeddings into LightGCN (Recall@20 **0.2047**).
- Diagnosed oversmoothing as the key failure mode and recovered up to **4.5%** Recall@20 with APPNP; documented in a technical report.

PROJECTS

Amazon ML Challenge 2026 – Business Entity Resolution *(Code)* *Python, LightGBM, scikit-learn, TF-IDF*

- Architected an end-to-end entity-resolution pipeline matching **1.73M** records against ~10M across three multilingual sources; scored **0.96 macro F0.5**.
- Reduced ~23 trillion possible comparisons to **92.1M** candidate pairs with TF-IDF blocking, keeping **94%** of true matches.
- Trained a LightGBM matcher (55 features, density-matched negatives), running the full pipeline in **~3 h on CPU** — no GPU.

PDF Chat with AI *(Code | Demo)* *FastAPI, React, LangChain, FAISS*

- Delivered a full-stack multi-turn RAG assistant (FastAPI + React) over uploaded PDFs, fusing FAISS-retrieved context with chat history in Gemini prompts.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, JavaScript
Core CS: DSA, Object-Oriented & Low-Level Design, DBMS, Operating Systems, Computer Networks
Backend & Databases: FastAPI, Flask, REST APIs, React, MySQL, SQLite
ML & GenAI: PyTorch, Scikit-learn, LightGBM, XGBoost, MLflow, LLMs, RAG, LangChain, FAISS, Sentence Transformers
Tools: Git, GitHub, Linux, Jupyter, Cloud Deployment (Render)

ACHIEVEMENTS

- Solved **1000+** DSA problems across LeetCode, GeeksforGeeks and CodeChef.
- Achieved a **LeetCode contest rating of 1600+** and **CodeChef 3-Star** (1600+ rating).
- Selected for the merit-based **IIT (ISM) Dhanbad Summer Research Internship 2026**; among ~90 interns institute-wide.
- Placed in the **top 1,000 of 30,000+ teams** in the **Amazon ML Challenge 2026** (Business Entity Resolution).
- Reached Round 2 of **Adobe University Hackathon 2026** as a team, in the top 20% of 300,000+ participants.